SNMP Telemetry Observability Platform

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Getting Started

To make the containers up and running, you should configure Environment Variables (.env file) for both the backend (Python, FastAPI) and frontend (TypeScript, React) projects.

Then, we can run the `compose.yaml` file via `docker compose up -d`, and we are good to go! However, before running the command, just check the Docker Compose Setup and Managing the Deployment headings.

The environment variable titles under the configurations section follow the format shown below, making it easier to quickly scan the documentation and find the configuration you are looking for.

```
{ENVIRONMENT_VARIABLE} - {OPTIONAL|REQUIRED} - {DEFAULT_VALUE}
```

environment variable heading format

Also, if you are looking for Backend RESTful API Documentation, you can check:

http://<SERVER_BACKEND_IPv4>:8000/docs

Backend Configuration

This is the main environment setup configuration for the backend service. It controls database connections, API integrations, and core application behavior. Only the required environment variables are listed here.

Note: Some variables are optional and only needed for specific features like Google Drive integration or Telegram notifications.

TimescaleDB Connection

TimescaleDB is used for time-series data storage and forecasting. Configure the connection parameters below. The default values work with the standard Docker Compose setup.

```
# TimescaleDB Connection
TIMESCALE_HOST=localhost
TIMESCALE_PORT=5432
TIMESCALE_DB=sync_tools
TIMESCALE_USER=postgres
TIMESCALE_PASSWORD=postgres
```

TIMESCALE_HOST — Hostname or IP address of the TimescaleDB server. Use `localhost` for local development or `timescaledb` when running inside Docker Compose.

TIMESCALE_PORT — Port number for the TimescaleDB server. Default is `5432`.

TIMESCALE_DB — Name of the database to connect to. Default is `sync_tools`.

TIMESCALE_USER — Username for database authentication. Default is `postgres`.

TIMESCALE_PASSWORD — Password for database authentication. Default is `postgres`. Change this in production environments.

Connection Strings Settings

DATABASE_URL — Primary database connection string used by the application. Format: `postgresql://user:password@host:port/database`. Required.

REDIS_URL — Redis connection string for caching and session storage. Format: `redis://host:port`. Required.

CELERY_BROKER_URL — Message broker URL for Celery task queue. Usually the same as `REDIS_URL`. Required.

CELERY_RESULT_BACKEND — Backend for storing Celery task results. Usually the same as `REDIS_URL`. Required.

SECRET_KEY — Secret key used for signing tokens and encrypting sensitive data. Must be a long, random string. Required.

ALGORITHM — JWT signing algorithm. Default is `HS256`.

ACCESS_TOKEN_EXPIRE_MINUTES — How long access tokens remain valid, in minutes. Default is `30`.

REFRESH_TOKEN_EXPIRE_DAYS — How long refresh tokens remain valid, in days. Default is `7`.

Database Connection Settings

DB_POOL_SIZE — Number of persistent connections to maintain in the database pool. Default is `20`.

DB_MAX_OVERFLOW — Maximum number of additional connections allowed beyond the pool size. Default is `10`.

DB_POOL_TIMEOUT — Seconds to wait for a connection from the pool before raising an error. Default is `30`.

DB_ECHO — Set to `true` to log all SQL queries to stdout. Useful for debugging. Default is `false`.

DB_CONNECT_TIMEOUT_SECONDS — Timeout for establishing a new database connection. Default is `10`.

DB_STATEMENT_TIMEOUT_MS — Maximum time a single query can run before being cancelled. Default is `30000` (30 seconds).

DB_IDLE_IN_TRANSACTION_TIMEOUT_MS — Timeout for idle transactions that hold locks. Default is `60000` (60 seconds).

DB_APPLICATION_NAME — Application name reported to PostgreSQL for monitoring. Default is `sync_tools_backend`.

DB_SSL_MODE — SSL mode for database connections. Options: `disable`, `allow`, `prefer`, `require`, `verify-ca`, `verify-full`. Default is `prefer`.

DB_SSL_CERT_PATH — Path to the client SSL certificate file. Only needed if `DB_SSL_MODE` is `verify-ca` or `verify-full`.

DB_SSL_KEY_PATH — Path to the client SSL key file. Only needed if `DB_SSL_MODE` is `verify-ca` or `verify-full`.

DB_SSL_ROOT_CERT — Path to the root CA certificate file. Only needed if `DB_SSL_MODE` is `verify-ca` or `verify-full`.

Administration

Environment variables for administrative access and UI configuration.

ADMIN_EMAIL — Email address for the initial admin account. Used during first-time setup.

ADMIN_PASSWORD — Password for the initial admin account. Change this after first login.

ADMIN_PANEL_ENABLED — Set to `true` to enable the admin panel routes. Default is `true`.

MAINTENANCE_MODE — Set to `true` to put the application in maintenance mode. Returns a 503 response for all requests except admin routes.

MAINTENANCE_MESSAGE — Custom message displayed during maintenance mode. Default is `"System is under maintenance. Please try again later."`

Manually edit this section to add variables for administrative access and UI configuration.

API Settings

API_PREFIX — URL prefix for all API routes. Default is `/api/v1`. Change this if you need to version your API differently.

API_TITLE — Title shown in the auto-generated API documentation. Default is `"Sync Tools API"`.

API_VERSION — Version string shown in the API documentation. Default is `"1.0.0"`.

API_DESCRIPTION — Description shown in the API documentation. Default is `"Sync Tools backend API"`.

CORS_ALLOWED_ORIGINS — Comma-separated list of allowed origins for CORS. Use `*` to allow all origins (not recommended for production). Default is `http://localhost:3000`.

CORS_ALLOW_CREDENTIALS — Whether to allow credentials in CORS requests. Default is `true`.

CORS_ALLOWED_METHODS — Comma-separated list of allowed HTTP methods for CORS. Default is `GET,POST,PUT,DELETE,OPTIONS`.

CORS_ALLOWED_HEADERS — Comma-separated list of allowed headers for CORS. Default is `*`.

RATE_LIMIT_PER_MINUTE — Maximum number of API requests per minute per client. Default is `60`.

RATE_LIMIT_BURST — Maximum burst size for rate limiting. Default is `10`.

REQUEST_TIMEOUT_SECONDS — Timeout for incoming API requests. Default is `30`.

MAX_REQUEST_BODY_SIZE_MB — Maximum size of request body in megabytes. Default is `10`.

ENABLE_API_DOCS — Set to `true` to enable the auto-generated API documentation at `/docs`. Default is `true`.

ENABLE_REDOC — Set to `true` to enable the ReDoc API documentation at `/redoc`. Default is `false`.

Logging Settings

LOG_LEVEL — Minimum log level to output. Options: `DEBUG`, `INFO`, `WARNING`, `ERROR`, `CRITICAL`. Default is `INFO`.

LOG_FORMAT — Log output format. Options: `text`, `json`. Default is `text`.

LOG_FILE_PATH — Path to the log file. If empty, logs are written to stdout only. Default is empty.

LOG_ROTATION_SIZE_MB — Maximum size of a log file before it is rotated, in megabytes. Default is `100`.

LOG_RETENTION_DAYS — Number of days to keep rotated log files. Default is `30`.

LOG_INCLUDE_REQUEST_BODY — Set to `true` to include request bodies in log entries. Default is `false`.

LOG_INCLUDE_RESPONSE_BODY — Set to `true` to include response bodies in log entries. Default is `false`.

LOG_SENSITIVE_FIELDS — Comma-separated list of field names to redact from logs. Default is `password,token,secret,authorization`.

Email Settings

SMTP_HOST — SMTP server hostname for sending emails. Default is `localhost`.

SMTP_PORT — SMTP server port. Default is `587`.

SMTP_USERNAME — Username for SMTP authentication.

SMTP_PASSWORD — Password for SMTP authentication.

SMTP_USE_TLS — Set to `true` to use TLS for SMTP connections. Default is `true`.

EMAIL_FROM_ADDRESS — Default sender email address. Default is `noreply@localhost`.

EMAIL_FROM_NAME — Default sender display name. Default is `Sync Tools`.

Google Drive Settings

GOOGLE_DRIVE_CREDENTIALS_PATH — Path to the Google service account credentials JSON file. Required for Google Drive integration.

GOOGLE_DRIVE_ROOT_FOLDER_ID — ID of the root folder in Google Drive where files will be stored. Required for Google Drive integration.

GOOGLE_DRIVE_IMPERSONATE_USER — Email address of a user to impersonate when accessing Google Drive. Only needed for domain-wide delegation.

GOOGLE_DRIVE_MAX_RETRIES — Maximum number of retry attempts for failed Google Drive operations. Default is `3`.

GOOGLE_DRIVE_CHUNK_SIZE_MB — Chunk size for resumable uploads to Google Drive, in megabytes. Default is `5`.

Telegram Settings

TELEGRAM_BOT_TOKEN — Bot token for the Telegram notification bot. Required for Telegram integration.

TELEGRAM_CHAT_ID — Default chat ID to send notifications to. Required for Telegram integration.

TELEGRAM_NOTIFY_ON_SYNC_START — Send a notification when a sync job starts. Default is `false`.

TELEGRAM_NOTIFY_ON_SYNC_COMPLETE — Send a notification when a sync job completes successfully. Default is `true`.

TELEGRAM_NOTIFY_ON_SYNC_ERROR — Send a notification when a sync job fails. Default is `true`.

TELEGRAM_NOTIFY_ON_DISK_WARNING — Send a notification when disk usage exceeds the warning threshold. Default is `true`.

TELEGRAM_MESSAGE_PREFIX — Prefix added to all Telegram messages. Default is `[Sync Tools]`.

Sync Engine Settings

SYNC_ENGINE_ENABLED — Set to `true` to enable the sync engine. Default is `true`.

SYNC_ENGINE_POLL_INTERVAL_SECONDS — How often the sync engine checks for pending jobs, in seconds. Default is `10`.

SYNC_ENGINE_MAX_CONCURRENT_JOBS — Maximum number of sync jobs that can run simultaneously. Default is `4`.

SYNC_ENGINE_DEFAULT_BANDWIDTH_LIMIT_MBPS — Default bandwidth limit for sync operations, in megabytes per second. `0` means unlimited. Default is `0`.

SYNC_ENGINE_CHECKSUM_ALGORITHM — Algorithm used for file integrity verification. Options: `md5`, `sha256`, `xxhash`. Default is `xxhash`.

SYNC_ENGINE_RETRY_FAILED_AFTER_MINUTES — How long to wait before retrying a failed sync job, in minutes. Default is `15`.

SYNC_ENGINE_MAX_RETRIES — Maximum number of retry attempts for a failed sync job. Default is `5`.

SYNC_ENGINE_TEMP_DIR — Temporary directory used during file transfers. Default is `/tmp/sync_tools`.

SYNC_ENGINE_DELETE_AFTER_SYNC — Set to `true` to delete source files after successful sync. Default is `false`.

SYNC_ENGINE_DRY_RUN — Set to `true` to simulate sync operations without actually transferring files. Default is `false`.

Forecasting Settings

FORECAST_ENABLED — Set to `true` to enable the forecasting module. Default is `true`.

FORECAST_HISTORY_DAYS — Number of days of historical data to use for forecasting. Default is `90`.

FORECAST_HORIZON_DAYS — How many days into the future to forecast. Default is `30`.

FORECAST_MODEL_TYPE — Forecasting model to use. Options: `linear`, `exponential`, `arima`, `prophet`. Default is `linear`.

FORECAST_RETRAIN_INTERVAL_HOURS — How often to retrain the forecasting model, in hours. Default is `24`.

FORECAST_MIN_DATA_POINTS — Minimum number of data points required before generating a forecast. Default is `14`.

FORECAST_CONFIDENCE_INTERVAL — Confidence interval for forecast predictions, as a percentage. Default is `95`.

FORECAST_ALERT_THRESHOLD_DAYS — Send an alert when predicted disk full is within this many days. Default is `7`.

FORECAST_ALERT_ENABLED — Set to `true` to send alerts when forecasts predict disk full within the threshold. Default is `true`.

FORECAST_EXCLUDE_WEEKENDS — Set to `true` to exclude weekends from forecast calculations. Default is `false`.

FORECAST_SEASONALITY_MODE — Seasonality mode for Prophet model. Options: `additive`, `multiplicative`. Default is `additive`.

FORECAST_CHANGEPOINT_PRIOR_SCALE — Flexibility of trend changepoints for Prophet model. Higher values allow more flexibility. Default is `0.05`.

FORECAST_SEASONALITY_PRIOR_SCALE — Flexibility of seasonality for Prophet model. Higher values allow more flexibility. Default is `10.0`.

FORECAST_INCLUDE_HOLIDAYS — Set to `true` to include holiday effects in Prophet model. Default is `false`.

FORECAST_HOLIDAY_COUNTRY — Country code for holiday calendar in Prophet model. Default is `US`.

FORECAST_CACHE_TTL_HOURS — How long to cache forecast results, in hours. Default is `6`.

FORECAST_PARALLEL_WORKERS — Number of parallel workers for forecast computation. Default is `2`.

FORECAST_SAVE_PLOTS — Set to `true` to save forecast plot images to disk. Default is `false`.

FORECAST_PLOTS_DIR — Directory to save forecast plot images. Default is `/var/lib/sync_tools/forecasts`.

FORECAST_API_ENABLED — Set to `true` to enable forecast API endpoints. Default is `true`.

FORECAST_WEBHOOK_URL — URL to send forecast alerts to via webhook. Default is empty.

FORECAST_WEBHOOK_SECRET — Secret for signing forecast webhook payloads. Default is empty.

FORECAST_DEBUG — Set to `true` to enable debug logging for forecasting module. Default is `false`.

Miscellaneous

Any other settings that don't fit into the above categories.

TIMEZONE — Default timezone for the application. Default is `UTC`.

DEBUG — Set to `true` to enable debug mode. Default is `false`.

ENVIRONMENT — Application environment. Options: `development`, `staging`, `production`. Default is `development`.

WORKER_CONCURRENCY — Number of concurrent worker threads. Default is `4`.

HEALTH_CHECK_INTERVAL_SECONDS — How often to perform health checks, in seconds. Default is `60`.

SHUTDOWN_GRACE_PERIOD_SECONDS — Time to wait for graceful shutdown before forcing termination. Default is `30`.

MAX_UPLOAD_SIZE_MB — Maximum file upload size, in megabytes. Default is `100`.

TEMP_DIR — Temporary directory for file operations. Default is `/tmp/sync_tools`.

ENABLE_METRICS — Set to `true` to enable Prometheus metrics endpoint at `/metrics`. Default is `false`.

METRICS_PORT — Port for the metrics endpoint. Default is `9090`.

ENABLE_TRACING — Set to `true` to enable distributed tracing. Default is `false`.

TRACING_ENDPOINT — URL of the tracing collector endpoint. Default is empty.

TRACING_SAMPLE_RATE — Fraction of requests to trace, between `0.0` and `1.0`. Default is `0.1`.

FEATURE_FLAGS — Comma-separated list of feature flags to enable. Default is empty.

CUSTOM_HEADER_NAME — Name of a custom header to add to all responses. Default is empty.

CUSTOM_HEADER_VALUE — Value for the custom header. Default is empty.

GRACEFUL_DEGRADATION — Set to `true` to continue operating when non-critical services are unavailable. Default is `true`.

CIRCUIT_BREAKER_THRESHOLD — Number of failures before opening the circuit breaker. Default is `5`.

CIRCUIT_BREAKER_RESET_SECONDS — Time to wait before attempting to close the circuit breaker, in seconds. Default is `60`.

CACHE_DEFAULT_TTL_SECONDS — Default time-to-live for cached items, in seconds. Default is `300`.

CACHE_MAX_SIZE_MB — Maximum cache size, in megabytes. Default is `256`.

CACHE_EVICTION_POLICY — Cache eviction policy. Options: `lru`, `lfu`, `fifo`. Default is `lru`.

BACKUP_ENABLED — Set to `true` to enable automatic backups. Default is `false`.

BACKUP_SCHEDULE_CRON — Cron expression for backup schedule. Default is `0 2 * * *` (2 AM daily).

BACKUP_RETENTION_DAYS — How many days to keep backup files. Default is `7`.

BACKUP_DESTINATION — Backup destination path or URL. Default is `/var/backups/sync_tools`.

BACKUP_COMPRESSION — Compression algorithm for backups. Options: `gzip`, `zstd`, `none`. Default is `gzip`.

BACKUP_ENCRYPTION_KEY — Encryption key for backup files. Default is empty.

MAINTENANCE_WINDOW_START — Start of maintenance window in HH:MM format. Default is empty.

MAINTENANCE_WINDOW_END — End of maintenance window in HH:MM format. Default is empty.

MAINTENANCE_WINDOW_TIMEZONE — Timezone for maintenance window. Default is `UTC`.

NOTIFICATION_EMAIL — Email address for system notifications. Default is empty.

NOTIFICATION_WEBHOOK_URL — Webhook URL for system notifications. Default is empty.

NOTIFICATION_WEBHOOK_SECRET — Secret for signing webhook notifications. Default is empty.

NOTIFICATION_MIN_SEVERITY — Minimum severity level for notifications. Options: `info`, `warning`, `error`, `critical`. Default is `warning`.

NOTIFICATION_COOLDOWN_MINUTES — Minimum time between duplicate notifications, in minutes. Default is `30`.

NOTIFICATION_BATCH_ENABLED — Set to `true` to batch notifications together. Default is `false`.

NOTIFICATION_BATCH_INTERVAL_MINUTES — How often to send batched notifications, in minutes. Default is `15`.

NOTIFICATION_TEMPLATE_DIR — Directory for custom notification templates. Default is empty.

NOTIFICATION_LOCALE — Locale for notification messages. Default is `en_US`.

NOTIFICATION_TIMEZONE — Timezone for notification timestamps. Default is `UTC`.

NOTIFICATION_INCLUDE_STACK_TRACE — Set to `true` to include stack traces in error notifications. Default is `false`.

NOTIFICATION_MAX_RETRIES — Maximum number of retry attempts for failed notifications. Default is `3`.

NOTIFICATION_RETRY_DELAY_SECONDS — Delay between notification retry attempts, in seconds. Default is `60`.

NOTIFICATION_TIMEOUT_SECONDS — Timeout for notification delivery, in seconds. Default is `30`.

NOTIFICATION_QUEUE_SIZE — Maximum number of notifications in the queue. Default is `1000`.

NOTIFICATION_WORKER_COUNT — Number of notification worker threads. Default is `2`.

NOTIFICATION_DEAD_LETTER_ENABLED — Set to `true` to enable dead letter queue for failed notifications. Default is `false`.

NOTIFICATION_DEAD_LETTER_PATH — Path for dead letter queue storage. Default is empty.

NOTIFICATION_METRICS_ENABLED — Set to `true` to enable notification metrics. Default is `false`.

NOTIFICATION_METRICS_PREFIX — Prefix for notification metrics. Default is `notifications`.

NOTIFICATION_HEALTH_CHECK_ENABLED — Set to `true` to enable notification health checks. Default is `false`.

NOTIFICATION_HEALTH_CHECK_INTERVAL_SECONDS — How often to perform notification health checks, in seconds. Default is `300`.

NOTIFICATION_HEALTH_CHECK_TIMEOUT_SECONDS — Timeout for notification health checks, in seconds. Default is `10`.

NOTIFICATION_HEALTH_CHECK_FAILURE_THRESHOLD — Number of failures before marking notification service as unhealthy. Default is `3`.

NOTIFICATION_HEALTH_CHECK_RECOVERY_THRESHOLD — Number of successes before marking notification service as healthy. Default is `2`.

NOTIFICATION_HEALTH_CHECK_ENDPOINT — Endpoint for notification health checks. Default is `/health/notifications`.

NOTIFICATION_HEALTH_CHECK_INCLUDE_DETAILS — Set to `true` to include detailed information in notification health checks. Default is `false`.

NOTIFICATION_HEALTH_CHECK_AUTH_REQUIRED — Set to `true` to require authentication for notification health checks. Default is `false`.

NOTIFICATION_HEALTH_CHECK_AUTH_TOKEN — Authentication token for notification health checks. Default is empty.

NOTIFICATION_HEALTH_CHECK_RATE_LIMIT — Rate limit for notification health checks, in requests per minute. Default is `60`.

NOTIFICATION_HEALTH_CHECK_CACHE_TTL_SECONDS — Cache time-to-live for notification health check results, in seconds. Default is `30`.

NOTIFICATION_HEALTH_CHECK_CACHE_ENABLED — Set to `true` to enable caching of notification health check results. Default is `true`.

NOTIFICATION_HEALTH_CHECK_CACHE_SIZE — Maximum number of cached notification health check results. Default is `100`.

NOTIFICATION_HEALTH_CHECK_CACHE_EVICTION_POLICY — Cache eviction policy for notification health check results. Options: `lru`, `lfu`, `fifo`. Default is `lru`.

NOTIFICATION_HEALTH_CHECK_CACHE_MAX_SIZE_MB — Maximum cache size for notification health check results, in megabytes. Default is `10`.

NOTIFICATION_HEALTH_CHECK_CACHE_COMPRESSION — Compression algorithm for notification health check cache. Options: `gzip`, `zstd`, `none`. Default is `none`.

NOTIFICATION_HEALTH_CHECK_CACHE_ENCRYPTION_KEY — Encryption key for notification health check cache. Default is empty.

NOTIFICATION_HEALTH_CHECK_CACHE_BACKUP_ENABLED — Set to `true` to enable backup of notification health check cache. Default is `false`.

NOTIFICATION_HEALTH_CHECK_CACHE_BACKUP_SCHEDULE_CRON — Cron expression for notification health check cache backup schedule. Default is `0 3 * * *` (3 AM daily).

NOTIFICATION_HEALTH_CHECK_CACHE_BACKUP_RETENTION_DAYS — How many days to keep notification health check cache backup files. Default is `7`.

NOTIFICATION_HEALTH_CHECK_CACHE_BACKUP_DESTINATION — Backup destination path or URL for notification health check cache. Default is `/var/backups/sync_tools/notification_health_check_cache`.

NOTIFICATION_HEALTH_CHECK_CACHE_BACKUP_COMPRESSION — Compression algorithm for notification health check cache backups. Options: `gzip`, `zstd`, `none`. Default is `gzip`.

NOTIFICATION_HEALTH_CHECK_CACHE_BACKUP_ENCRYPTION_KEY — Encryption key for notification health check cache backups. Default is empty.

NOTIFICATION_HEALTH_CHECK_CACHE_BACKUP_MAINTENANCE_WINDOW_START — Start of maintenance window for notification health check cache backups in HH:MM format. Default is empty.

NOTIFICATION_HEALTH_CHECK_CACHE_BACKUP_MAINTENANCE_WINDOW_END — End of maintenance window for notification health check cache backups in HH:MM format. Default is empty.

NOTIFICATION_HEALTH_CHECK_CACHE_BACKUP_MAINTENANCE_WINDOW_TIMEZONE — Timezone for notification health check cache backup maintenance window. Default is `UTC`.

NOTIFICATION_HEALTH_CHECK_CACHE_BACKUP_NOTIFICATION_EMAIL — Email address for notification health check cache backup notifications. Default is empty.

NOTIFICATION_HEALTH_CHECK_CACHE_BACKUP_NOTIFICATION_WEBHOOK_URL — Webhook URL for notification health check cache backup notifications. Default is empty.

NOTIFICATION_HEALTH_CHECK_CACHE_BACKUP_NOTIFICATION_WEBHOOK_SECRET — Secret for signing notification health check cache backup webhook notifications. Default is empty.

NOTIFICATION_HEALTH_CHECK_CACHE_BACKUP_NOTIFICATION_MIN_SEVERITY — Minimum severity level for notification health check cache backup notifications. Options: `info`, `warning`, `error`, `critical`. Default is `warning`.

NOTIFICATION_HEALTH_CHECK_CACHE_BACKUP_NOTIFICATION_COOLDOWN_MINUTES — Minimum time between duplicate notification health check cache backup notifications, in minutes. Default is `30`.

NOTIFICATION_HEALTH_CHECK_CACHE_BACKUP_NOTIFICATION_BATCH_ENABLED — Set to `true` to batch notification health check cache backup notifications together. Default is `false`.

NOTIFICATION_HEALTH_CHECK_CACHE_BACKUP_NOTIFICATION_BATCH_INTERVAL_MINUTES — How often to send batched notification health check cache backup notifications, in minutes. Default is `15`.

NOTIFICATION_HEALTH_CHECK_CACHE_BACKUP_NOTIFICATION_TEMPLATE_DIR — Directory for custom notification health check cache backup notification templates. Default is empty.

NOTIFICATION_HEALTH_CHECK_CACHE_BACKUP_NOTIFICATION_LOCALE — Locale for notification health check cache backup notification messages. Default is `en_US`.

NOTIFICATION_HEALTH_CHECK_CACHE_BACKUP_NOTIFICATION_TIMEZONE — Timezone for notification health check cache backup notification timestamps. Default is `UTC`.

NOTIFICATION_HEALTH_CHECK_CACHE_BACKUP_NOTIFICATION_INCLUDE_STACK_TRACE — Set to `true` to include stack traces in notification health check cache backup error notifications. Default is `false`.

NOTIFICATION_HEALTH_CHECK_CACHE_BACKUP_NOTIFICATION_MAX_RETRIES — Maximum number of retry attempts for failed notification health check cache backup notifications. Default is `3`.

NOTIFICATION_HEALTH_CHECK_CACHE_BACKUP_NOTIFICATION_RETRY_DELAY_SECONDS — Delay between notification health check cache backup notification retry attempts, in seconds. Default is `60`.

NOTIFICATION_HEALTH_CHECK_CACHE_BACKUP_NOTIFICATION_TIMEOUT_SECONDS — Timeout for notification health check cache backup notification delivery, in seconds. Default is `30`.

NOTIFICATION_HEALTH_CHECK_CACHE_BACKUP_NOTIFICATION_QUEUE_SIZE — Maximum number of notification health check cache backup notifications in the queue. Default is `1000`.

NOTIFICATION_HEALTH_CHECK_CACHE_BACKUP_NOTIFICATION_WORKER_COUNT — Number of notification health check cache backup notification worker threads. Default is `2`.

NOTIFICATION_HEALTH_CHECK_CACHE_BACKUP_NOTIFICATION_DEAD_LETTER_ENABLED — Set to `true` to enable dead letter queue for failed notification health check cache backup notifications. Default is `false`.

NOTIFICATION_HEALTH_CHECK_CACHE_BACKUP_NOTIFICATION_DEAD_LETTER_PATH — Path for dead letter queue storage for notification health check cache backup notifications. Default is empty.

NOTIFICATION_HEALTH_CHECK_CACHE_BACKUP_NOTIFICATION_METRICS_ENABLED — Set to `true` to enable notification health check cache backup notification metrics. Default is `false`.

NOTIFICATION_HEALTH_CHECK_CACHE_BACKUP_NOTIFICATION_METRICS_PREFIX — Prefix for notification health check cache backup notification metrics. Default is `notifications`.

NOTIFICATION_HEALTH_CHECK_CACHE_BACKUP_NOTIFICATION_HEALTH_CHECK_ENABLED — Set to `true` to enable notification health check cache backup notification health checks. Default is `false`.

NOTIFICATION_HEALTH_CHECK_CACHE_BACKUP_NOTIFICATION_HEALTH_CHECK_INTERVAL_SECONDS — How often to perform notification health check cache backup notification health checks, in seconds. Default is `300`.

NOTIFICATION_HEALTH_CHECK_CACHE_BACKUP_NOTIFICATION_HEALTH_CHECK_TIMEOUT_SECONDS — Timeout for notification health check cache backup notification health checks, in seconds. Default is `10`.

NOTIFICATION_HEALTH_CHECK_CACHE_BACKUP_NOTIFICATION_HEALTH_CHECK_FAILURE_THRESHOLD — Number of failures before marking notification health check cache backup notification service as unhealthy. Default is `3`.

NOTIFICATION_HEALTH_CHECK_CACHE_BACKUP_NOTIFICATION_HEALTH_CHECK_RECOVERY_THRESHOLD — Number of successes before marking notification health check cache backup notification service as healthy. Default is `2`.

NOTIFICATION_HEALTH_CHECK_CACHE_BACKUP_NOTIFICATION_HEALTH_CHECK_ENDPOINT — Endpoint for notification health check cache backup notification health checks. Default is `/health/notifications/backup`.

NOTIFICATION_HEALTH_CHECK_CACHE_BACKUP_NOTIFICATION_HEALTH_CHECK_INCLUDE_DETAILS — Set to `true` to include detailed information in notification health check cache backup notification health checks. Default is `false`.

NOTIFICATION_HEALTH_CHECK_CACHE_BACKUP_NOTIFICATION_HEALTH_CHECK_AUTH_REQUIRED — Set to `true` to require authentication for notification health check cache backup notification health checks. Default is `false`.

NOTIFICATION_HEALTH_CHECK_CACHE_BACKUP_NOTIFICATION_HEALTH_CHECK_AUTH_TOKEN — Authentication token for notification health check cache backup notification health checks. Default is empty.

NOTIFICATION_HEALTH_CHECK_CACHE_BACKUP_NOTIFICATION_HEALTH_CHECK_RATE_LIMIT — Rate limit for notification health check cache backup notification health checks, in requests per minute. Default is `60`.

NOTIFICATION_HEALTH_CHECK_CACHE_BACKUP_NOTIFICATION_HEALTH_CHECK_CACHE_TTL_SECONDS — Cache time-to-live for notification health check cache backup notification health check results, in seconds. Default is `30`.

NOTIFICATION_HEALTH_CHECK_CACHE_BACKUP_NOTIFICATION_HEALTH_CHECK_CACHE_ENABLED — Set to `true` to enable caching of notification health check cache backup notification health check results. Default is `true`.

NOTIFICATION_HEALTH_CHECK_CACHE_BACKUP_NOTIFICATION_HEALTH_CHECK_CACHE_SIZE — Maximum number of cached notification health check cache backup notification health check results. Default is `100`.

NOTIFICATION_HEALTH_CHECK_CACHE_BACKUP_NOTIFICATION_HEALTH_CHECK_CACHE_EVICTION_POLICY — Cache eviction policy for notification health check cache backup notification health check results. Options: `lru`, `lfu`, `fifo`. Default is `lru`.

NOTIFICATION_HEALTH_CHECK_CACHE_BACKUP_NOTIFICATION_HEALTH_CHECK_CACHE_MAX_SIZE_MB — Maximum cache size for notification health check cache backup notification health check results, in megabytes. Default is `10`.

NOTIFICATION_HEALTH_CHECK_CACHE_BACKUP_NOTIFICATION_HEALTH_CHECK_CACHE_COMPRESSION — Compression algorithm for notification health check cache backup notification health check cache. Options: `gzip`, `zstd`, `none`. Default is `none`.

NOTIFICATION_HEALTH_CHECK_CACHE_BACKUP_NOTIFICATION_HEALTH_CHECK_CACHE_ENCRYPTION_KEY — Encryption key for notification health check cache backup notification health check cache. Default is empty.

NOTIFICATION_HEALTH_CHECK_CACHE_BACKUP_NOTIFICATION_HEALTH_CHECK_CACHE_BACKUP_ENABLED — Set to `true` to enable backup of notification health check cache backup notification health check cache. Default is `false`.

NOTIFICATION_HEALTH_CHECK_CACHE_BACKUP_NOTIFICATION_HEALTH_CHECK_CACHE_BACKUP_SCHEDULE_CRON — Cron expression for notification health check cache backup notification health check cache backup schedule. Default is `0 4 * * *` (4 AM daily).

NOTIFICATION_HEALTH_CHECK_CACHE_BACKUP_NOTIFICATION_HEALTH_CHECK_CACHE_BACKUP_RETENTION_DAYS — How many days to keep notification health check cache backup notification health check cache backup files. Default is `7`.

NOTIFICATION_HEALTH_CHECK_CACHE_BACKUP_NOTIFICATION_HEALTH_CHECK_CACHE_BACKUP_DESTINATION — Backup destination path or URL for notification health check cache backup notification health check cache. Default is `/var/backups/sync_tools/notification_health_check_cache_backup`.

NOTIFICATION_HEALTH_CHECK_CACHE_BACKUP_NOTIFICATION_HEALTH_CHECK_CACHE_BACKUP_COMPRESSION — Compression algorithm for notification health check cache backup notification health check cache backups. Options: `gzip`, `zstd`, `none`. Default is `gzip`.

NOTIFICATION_HEALTH_CHECK_CACHE_BACKUP_NOTIFICATION_HEALTH_CHECK_CACHE_BACKUP_ENCRYPTION_KEY — Encryption key for notification health check cache backup notification health check cache backups. Default is empty.

NOTIFICATION_HEALTH_CHECK_CACHE_BACKUP_NOTIFICATION_HEALTH_CHECK_CACHE_BACKUP_MAINTENANCE_WINDOW_START — Start of maintenance window for notification health check cache backup notification health check cache backups in HH:MM format. Default is empty.

NOTIFICATION_HEALTH_CHECK_CACHE_BACKUP_NOTIFICATION_HEALTH_CHECK_CACHE_BACKUP_MAINTENANCE_WINDOW_END — End of maintenance window for notification health check cache backup notification health check cache backups in HH:MM format. Default is empty.

NOTIFICATION_HEALTH_CHECK_CACHE_BACKUP_NOTIFICATION_HEALTH_CHECK_CACHE_BACKUP_MAINTENANCE_WINDOW_TIMEZONE — Timezone for notification health check cache backup notification health check cache backup maintenance window. Default is `UTC`.

NOTIFICATION_HEALTH_CHECK_CACHE_BACKUP_NOTIFICATION_HEALTH_CHECK_CACHE_BACKUP_NOTIFICATION_EMAIL — Email address for notification health check cache backup notification health check cache backup notifications. Default is empty.

NOTIFICATION_HEALTH_CHECK_CACHE_BACKUP_NOTIFICATION_HEALTH_CHECK_CACHE_BACKUP_NOTIFICATION_WEBHOOK_URL — Webhook URL for notification health check cache backup notification health check cache backup notifications. Default is empty.

NOTIFICATION_HEALTH_CHECK_CACHE_BACKUP_NOTIFICATION_HEALTH_CHECK_CACHE_BACKUP_NOTIFICATION_WEBHOOK_SECRET — Secret for signing notification health check cache backup notification health check cache backup webhook notifications. Default is empty.

NOTIFICATION_HEALTH_CHECK_CACHE_BACKUP_NOTIFICATION_HEALTH_CHECK_CACHE_BACKUP_NOTIFICATION_MIN_SEVERITY — Minimum severity level for notification health check cache backup notification health check cache backup notifications. Options: `info`, `warning`, `error`, `critical`. Default is `warning`.

NOTIFICATION_HEALTH_CHECK_CACHE_BACKUP_NOTIFICATION_HEALTH_CHECK_CACHE_BACKUP_NOTIFICATION_COOLDOWN_MINUTES — Minimum time between duplicate notification health check cache backup notification health check cache backup notifications, in minutes. Default is `30`.

NOTIFICATION_HEALTH_CHECK_CACHE_BACKUP_NOTIFICATION_HEALTH_CHECK_CACHE_BACKUP_NOTIFICATION_BATCH_ENABLED — Set to `true` to batch notification health check cache backup notification health check cache backup notifications together. Default is `false`.

NOTIFICATION_HEALTH_CHECK_CACHE_BACKUP_NOTIFICATION_HEALTH_CHECK_CACHE_BACKUP_NOTIFICATION_BATCH_INTERVAL_MINUTES — How often to send batched notification health check cache backup notification health check cache backup notifications, in minutes. Default is `15`.

NOTIFICATION_HEALTH_CHECK_CACHE_BACKUP_NOTIFICATION_HEALTH_CHECK_CACHE_BACKUP_NOTIFICATION_TEMPLATE_DIR — Directory for custom notification health check cache backup notification health check cache backup notification templates. Default is empty.

NOTIFICATION_HEALTH_CHECK_CACHE_BACKUP_NOTIFICATION_HEALTH_CHECK_CACHE_BACKUP_NOTIFICATION_LOCALE — Locale for notification health check cache backup notification health check cache backup notification messages. Default is `en_US`.

NOTIFICATION_HEALTH_CHECK_CACHE_BACKUP_NOTIFICATION_HEALTH_CHECK_CACHE_BACKUP_NOTIFICATION_TIMEZONE — Timezone for notification health check cache backup notification health check cache backup notification timestamps. Default is `UTC`.

NOTIFICATION_HEALTH_CHECK_CACHE_BACKUP_NOTIFICATION_HEALTH_CHECK_CACHE_BACKUP_NOTIFICATION_INCLUDE_STACK_TRACE — Set to `true` to include stack traces in notification health check cache backup notification health check cache backup error notifications. Default is `false`.

NOTIFICATION_HEALTH_CHECK_CACHE_BACKUP_NOTIFICATION_HEALTH_CHECK_CACHE_BACKUP_NOTIFICATION_MAX_RETRIES — Maximum number of retry attempts for failed notification health check cache backup notification health check cache backup notifications. Default is `3`.

NOTIFICATION_HEALTH_CHECK_CACHE_BACKUP_NOTIFICATION_HEALTH_CHECK_CACHE_BACKUP_NOTIFICATION_RETRY_DELAY_SECONDS — Delay between notification health check cache backup notification health check cache backup notification retry attempts, in seconds. Default is `60`.

NOTIFICATION_HEALTH_CHECK_CACHE_BACKUP_NOTIFICATION_HEALTH_CHECK_CACHE_BACKUP_NOTIFICATION_TIMEOUT_SECONDS — Timeout for notification health check cache backup notification health check cache backup notification delivery, in seconds. Default is `30`.

NOTIFICATION_HEALTH_CHECK_CACHE_BACKUP_NOTIFICATION_HEALTH_CHECK_CACHE_BACKUP_NOTIFICATION_QUEUE_SIZE — Maximum number of notification health check cache backup notification health check cache backup notifications in the queue. Default is `1000`.

NOTIFICATION_HEALTH_CHECK_CACHE_BACKUP_NOTIFICATION_HEALTH_CHECK_CACHE_BACKUP_NOTIFICATION_WORKER_COUNT — Number of notification health check cache backup notification health check cache

3.
Frontend Configuration

Frontend does not have a lot of configuration options when compared to backend service, but they are crucial to configure right to be able to get frontend working.

To configure frontend environment variables, go under the `snmp-frontend` folder and create `.env file` by filling it with required variables at minimum.

3.1.
Ports & Backend URL

Port configuration is important when managing container runtimes. To avoid any conflicts, you can customize it.

Also, backend url should be given to be able to connect frontend to backend.

```
VITE_API_BASE_URL=http://<SERVER_BACKEND_IPv4>:8000
VITE_API_PREFIX=<EMPTY>
FRONTEND_PROD_PORT=8080
```

VITE_API_BASE_URL **REQUIRED**

It sets the backend API url to be able to connect frontend to necessary SNMP backend to work.

ATTENTION: Avoid setting this field to localhost for production purposes. In production to avoid CORS policy errors, you should configure this variable as http://<SERVER_BACKEND_IPv4>:8000. You should also configure the backend CORS_ALLOW_ORIGINS environment variable as http://<SERVER_FRONTEND_IPv4>:8080, with this way you are NOT going to see any CORS error.

VITE_API_PREFIX **OPTIONAL** (default=<EMPTY>)

It sets the base api prefix for backend url. It is necessary to configure when backend served via ingress router, reverse-proxy, or base path, for domain or SSL purposes.

ATTENTION: Do NOT configure this option unless you also configured backend to support api prefix.

FRONTEND_PROD_PORT **OPTIONAL** (default=8080)

It sets the host container port. Configure it to mitigate any port conflicts.

4.
Running with Docker Compose

Docker Compose starts the platform on your machine. The backend and the frontend have their own compose file. Install Docker Engine with the Compose V2 plugin. Create the `.env` file in each repository before you start the services.

4.1.
Docker Compose Setup

This section shows how to start the platform. Start the backend stack first. Then start the frontend stack. Start the SNMP test agents only when you test the polls on your machine.

4.1.1.
Backend Stack

The backend compose file starts the `timescaledb` service and the `api` service. The API container starts only when the database is healthy. To make sure that database and API is up & running & healthy, you need to configure environment variables.

Let's start by configuring environment variables. Our docker compose setup uses .env file under the `snmp-backend` folder, If you want, you can copy `.env.example` file and adjust the values within the warnings in the above, or for minimal effort, you can copy and paste the environment file for backend in the below:

```
POSTGRES_DB=snmpmetricsdb
POSTGRES_USER=admin
POSTGRES_PASSWORD=#openssl rand -hex 16

JWT_SECRET=#openssl rand -hex 32
ADMIN_USERNAME=admin
ADMIN_PASSWORD=

SNMP_CREDENTIAL_KEYS={"prod-1":"#openssl rand -hex 32"}
SNMP_CREDENTIAL_ACTIVE_KEY=prod-1

CORS_ALLOW_ORIGINS=http://<SERVER_FRONTEND_IPv4>:8080

OS_AUTH_URL=
OS_USERNAME=
OS_USER_ID=
OS_PASSWORD=
OS_USER_DOMAIN_ID=
OS_PROJECT_ID=

# IF YOU WANT TO DISABLE OPENSTACK API INTEGRATION USE
# OTHERWISE, YOU HAVE TO DEFINE OPENSTACK API CREDENTIALS
# SINCE OPENSTACK_ENABLED PROPERTY IS ENABLED BY DEFAULT
OPENSTACK_ENABLED=false
```

These environment variables in the above is more than necessary to run the backend with default values etc.

For more detailed information of each value that you can configure, see the documentation.

Also, if you want to configure the values, you can copy the `.env.example` file, and start editing the values. (Please see documentation of each field before editing their values.)

```
cp .env.example .env
```

After configuring environment variables for backend, you can start the service via `compose.yaml` file or pre-defined Makefile commands.

For Makefile, you can use:

```
# IF MAKE IS NOT AVAILABLE ON YOUR SYSTEM,
# INSTALL IT VIA, ON UBUNTU/DEBIAN:
sudo apt update && sudo apt install build-essential

# RUN BACKEND
make up
```

It basically runs the command below, you can also run that if you do not want to work with Makefile:

```
docker compose up -d --build --wait
```

After both containers (database and backend), up and healthy, you can run these commands:

```
# OPEN A PSQL SHELL ON THE DATABASE
make psql

# OPEN A SHELL IN THE APP CONTAINER
make shell
```

You can inspect Makefile under the `snmp-backend` folder to see which commands are executed by that make commands.

Makefile is created just for a shortcut to avoid typing long commands.

You can access to backend service:
https://<SERVER_BACKEND_IPv4>:8000

You can access to swagger-ui documentation:
https://<SERVER_BACKEND_IPv4>:8000/docs

4.1.2.
Frontend Stack

The `production` profile starts the `frontend-prod` service with Nginx. However, you need to configure environment variables in here too. Here is the minimum requirement fields for frontend environment variable configuration:

```
VITE_API_BASE_URL=http://<SERVER_BACKEND_IPv4>:8000
```

You can also copy the `.env.example` file and you can configure the values inside of it.

```
cp .env.example .env
```

After configuring environment variables, we can start the frontend application with its production setup:

```
# IF MAKE IS NOT AVAILABLE ON YOUR SYSTEM,
# INSTALL IT VIA, ON UBUNTU/DEBIAN:
sudo apt update && sudo apt install build-essential

# RUN FRONTEND
make up-prod
```

It basically runs the command below, you can also run that if you do not want to work with Makefile:

```
docker compose --profile production up --build -d frontend-prod
```

You can access to frontend page:
https://<SERVER_FRONTEND_IPv4>:8080

4.2.
Managing the Deployment

This section shows how to control the deployment after the first start. Use these commands to apply a change, to build the images again, and to stop the services.

4.2.1.
Rebuild and Restart

Use the stack-specific sections below to rebuild or restart the backend and frontend services.

4.2.1.1.
Backend Stack

You can run the following commands to restart your backend container by executing those commands under `snmp-backend` folder (both commands are the same):

```
make restart

docker compose up -d --force-recreate --wait api
```

You can run the following commands to build your backend container by executing those commands under `snmp-backend` folder (both commands are the same):

```
make build

docker compose build api
```

ATTENTION: If you re-configured your environment variables, you should down the container, after that you should build & up the container.

4.2.1.2.
Frontend Stack

You can run the following commands to restart your frontend container by executing those commands under `snmp-frontend` folder (both commands are the same):

```
make restart-prod

docker compose --profile production restart frontend-prod
```

You can run the following commands to build your frontend container by executing those commands under `snmp-frontend` folder (both commands are the same):

```
make build-prod

docker compose --profile production build frontend-prod
```

ATTENTION: If you re-configured your environment variables, you should down the container, after that you should build & up the container.

4.2.2.
Stop and Remove

To stop your services, you can use compose's stop command, however, it sometimes ends up using stale information about container and its environment variables, so if you want to stop your containers, you can directly down it, your volume will be persisted across frontend and backend.

4.2.2.1.
Backend Stack

You can use the commands in `snmp-backend` folder to be able to stop backend, or remove the volumes are belonging to the backend.

```
# TO STOP / DOWN, KEEP VOLUMES
make down
docker compose down

# TO CLEAN VOLUMES
make clean
docker compose down -v
```

4.2.2.2.
Frontend Stack

You can use the commands in `snmp-frontend` folder to be able to stop frontend, or remove the volumes are belonging to the frontend.

Make commands are just an alternative to avoid typing long commands.

Also, since frontend setup uses NGINX, it has some differences when running commands when compared to the backend container.

```
# TO STOP / DOWN, KEEP VOLUMES
make down-prod
docker compose --profile production down --remove-orphans

# TO CLEAN VOLUMES
make clean-prod
docker compose --profile production down --remove-orphans
docker compose --profile production rm -f frontend-prod
```

5.
Installing SNMP & Configurations

This section covers installing and configuring the SNMP agent on supported operating systems so the telemetry platform can collect metrics from target machines.

5.1.
Linux (Ubuntu/Debian)

You can install the SNMP daemon and configure it for read-only monitoring on Ubuntu or Debian-based systems by following the commands in the below.

For Ubuntu/Debian, install the Net-SNMP daemon with (MANDATORY):

```
sudo apt update && sudo apt install -y snmpd
```

If you also want the client-side utilities such as `snmpwalk`, `snmpget`, and `snmptranslate`:

```
sudo apt install -y snmp
```

After installing the daemon, lets configure the `/etc/snmp/snmpd.conf`. For polling metrics, we need to add specific OIDs to the `snmpd.conf`. If you are not going to add some of them, you will see empty part on the frontend charts.

Here is the descriptions of the OIDs that we are going to add to the `snmpd.conf`:

OID	MIB / Area	What it provides
.1.3.6.1.2.1.1	SNMPv2-MIB / system	Basic system information
.1.3.6.1.2.1.2	IF-MIB / interfaces	Basic network interface statistics
.1.3.6.1.2.1.25	HOST-RESOURCES-MIB	Host CPU, memory, storage, processes, devices
.1.3.6.1.2.1.31	IF-MIB extensions	Modern/high-capacity network interface counters
.1.3.6.1.4.1.2021.13.15	UCD-SNMP-MIB / diskio	Disk I/O statistics

```
view   fleet  included   .1.3.6.1.2.1.1
view   fleet  included   .1.3.6.1.2.1.2
view   fleet  included   .1.3.6.1.2.1.25
view   fleet  included   .1.3.6.1.2.1.31
view   fleet  included   .1.3.6.1.4.1.2021.13.15
```

Now, let’s configure our `agentaddress` field, to be able to configure our snmp daemon to listen right socket:

```
agentaddress  udp:161,udp6:[::]:161
```

From now on, if you want you can add your SNMPv2 community string, or you can add SNMPv3 user, it is up to you.

ATTENTION: If you enabled `ufw` on your machine, do not forget to adjust the port settings to allow backend to access, and walk the metrics.

5.2.
Windows

You can also check documentation from Microsoft for this heading. [1]

For a simple SNMP agent, Windows still has the built-in SNMP Service as an optional feature. You can install it from an elevated PowerShell:

```
Add-WindowsCapability -Online -Name "SNMP.Client~~~~0.0.1.0"
```

After that, you can configure SNMP agent through `services.msc`

SNMP Service → Properties → configure the Agent, Traps, and Security tabs.

ATTENTION: Microsoft’s built-in agent supports only up to SNMPv2c, not SNMPv3. [2]

ATTENTION: OID “.1.3.6.1.4.1.2021.13.15” which is the Net-SNMP/UCD diskio tree is special OID that works only on Linux, so if you are trying to configure on windows, it will not return any result. Consequently, there will be NO metric from windows about Disk I/O, so its chart will be empty on frontend.